

Week 1 Plan Explanation – Setup & Market Data Foundations

Week Goal

Set up the development environment, build a reliable data pipeline, and develop intuition for market data.

By the end of the week, the intern can build a reproducible research environment, validate OHLCV data, compute returns correctly, and produce defensible portfolio summaries from clean inputs.

Task 1: Onboarding & Environment

This task establishes the reproducible environment and working conventions that support every later analysis.

Detailed Work

- Set up Python environment (conda/venv), install dependencies
- Clone repo, understand folder structure and naming conventions
- Review Git workflow: feature branches, commit messages, pull requests
- Code style agreement: PEP 8, docstrings, type hints for public functions

Concrete Output

- Create `requirements.txt`, verify all packages install cleanly
- Write a "Hello World" notebook that imports all required libraries

Quality Check

A fresh clone must install successfully from `requirements.txt`, and the starter notebook must run from top to bottom without manual fixes.

Task 2: Data Loading & Initial Exploration

This task turns the raw OHLCV files into a clearly understood research dataset with an explicit schema and coverage profile.

Detailed Work

- Load raw OHLCV CSV/parquet files into pandas DataFrames
- Understand data schema: date, ticker, open, high, low, close, volume, adj_close
- Explore data dimensions: how many stocks, date range, frequency

Concrete Output

- Write `src/data_utils.py` with `load_ohlc()` function
- Print summary statistics for the full dataset (count, mean, std, min, max)
- Identify the top 10 and bottom 10 stocks by average daily volume

Quality Check

The loader must preserve adjusted prices, report universe and date coverage, and flag schema or trading–calendar inconsistencies.

Task 3: Data Quality & Cleaning

This task identifies data defects before they can contaminate return calculations, factors, or portfolio results.

Detailed Work

- Check for missing values: which stocks, which dates, patterns
- Handle missing data: forward fill, drop, interpolation – discuss trade-offs
- Detect outliers: returns $> \pm 50\%$ in a single day, volume spikes
- Check for survivorship bias awareness (which stocks enter/exit the dataset)

Concrete Output

- Build a `clean_ohlc()` function that handles NaN, adjusts for splits if needed

- Create a data quality report: % missing by year, list of suspicious tickers

Quality Check

Every cleaning rule must be documented; suspicious prices, missing observations, splits, and survivorship concerns must be surfaced rather than silently removed.

Task 4: Returns Computation

This task builds the return series used throughout the internship and clarifies how arithmetic and log returns differ.

Detailed Work

- Compute simple (arithmetic) returns: $R = (P_t - P_{t-1}) / P_{t-1}$
- Compute log returns: $r = \ln(P_t / P_{t-1})$
- Discuss when to use each: log returns for time aggregation, simple returns for cross-sectional aggregation
- Compute daily, weekly, monthly returns

Concrete Output

- Add `compute_returns()` to `data_utils.py` supporting both return types
- Plot return distributions for 5 stocks – are they normal? Compute skewness and kurtosis
- Compare daily vs monthly return volatility – verify $\sqrt{T}$ rule approximately holds

Quality Check

Return type and frequency must be explicit, weekly and monthly simple returns must compound correctly, and spot checks must reproduce manual calculations.

Task 5: Rolling Statistics & Portfolio Returns

This task connects cleaned security data to rolling diagnostics and the first investable portfolio comparisons.

Detailed Work

- Compute rolling mean, rolling std (20-day, 60-day, 252-day windows)
- Compute annualized volatility
- Equal-weight portfolio return vs value-weight portfolio return

Concrete Output

- Plot rolling 60-day volatility for SPY or market proxy
- Construct equal-weight and value-weight portfolios of 50 stocks, plot cumulative returns
- Compute and compare annualized return, volatility, and Sharpe ratio for both portfolios

Quality Check

Annualization assumptions and portfolio weights must be stated consistently, with cumulative return and rolling-risk plots reconciled to the summary table.

Week Deliverables

- `notebooks/week1/data_quality_checks.ipynb` – fully documented
- `src/data_utils.py` – load, clean, compute returns
- Weekly memo (1 page): data issues found, cleaning decisions made, key observations

Recommended Reading

- *Quantitative Equity Portfolio Management* (Chincarini & Kim), Chapter 1–2
- pandas documentation: MultiIndex, groupby, resample
- Investopedia: OHLCV, adjusted close, stock splits

Common Pitfalls

- Using unadjusted close prices (splits/dividends distort returns)
- Look-ahead bias: using future data in calculations
- Confusing log returns and simple returns when aggregating across stocks

What Success Looks Like

The intern can build a reproducible research environment, validate OHLCV data, compute returns correctly, and produce defensible portfolio summaries from clean inputs.